Preschool & Early Interventions: Perry, Abecedarian, and Fade-Out at Scale

Two small, high-fidelity 1960s-70s RCTs (Perry, Abecedarian) show large, persistent adult gains; modern at-scale preschool interventions often show cognitive fade-out.

ODERATE-HISTORICAL; LOW-GENERALIZATION | STATUS: DRAFT | DOMAIN: EARLY-LEARNING, COGNITION, POLICY | REPLICATION

WHAT WE KNOW (Narrowest Supported Conclusion)

Two small but rigorously randomized 1960s-70s preschool programs for disadvantaged children (Perry Preschool, Abecedarian) show sustained gains in adult education, employment, earnings, and health, and reduced crime, decades after the intervention ended -- but these are small, intensive, high-fidelity, historically-specific programs, not evidence that any preschool program at scale today replicates these effects.

EVIDENCE AND CAUSAL ASSESSMENT

- Perry Preschool and Abecedarian were both RCTs, each serving roughly 60 children, an unusually strong causal design for early-education research given the era.

- Long-run follow-up (Garcia et al., 2021) found increased adult education, employment, earnings, and health, and lower crime/incarceration in treatment vs. control groups; estimated benefits totaled six-to-ten times program costs.

- Reading/math gains persisted or grew through second grade and were maintained to ages 14-21 in Abecedarian follow-ups (Campbell et al., 2012).

- Perry Preschool evidence also suggests intergenerational spillover: better family environments and outcomes for children of the original participants (Garcia et al., 2023).

- A separate synthesis raises the question of whether modern, larger-scale preschool programs are becoming less effective than these historical small-N demonstration programs, implying scale/fidelity dependence.

EFFECT SIZE AND PERSISTENCE

Effects are large in the original small samples and unusually persistent -- among the longest-followed early-childhood RCTs in the literature (into the 4th-5th decade of life).

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: n/a (historical randomized controlled educational trials, not an observational parenting phenotype).

Small sample sizes (~60 per program) limit precision and generalizability; both programs were unusually intensive, high-fidelity, researcher-designed model programs, not typical at-scale public preschool. Benefits are plausibly tied to program quality/intensity/fidelity rather than preschool as a generic category.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not establish that any preschool program, or preschool attendance in general, produces these effect sizes; does not license extrapolating a specific dollar-benefit multiplier without similar fidelity and intensity.

WHAT ACTION IS REASONABLY JUSTIFIED

Institutional investment in high-fidelity, well-resourced early-childhood programs for disadvantaged children is supported by unusually strong, long-run causal evidence from these two programs; assuming any preschool expansion will replicate Perry/Abecedarian-scale effects is not justified without attention to implementation quality.

VERIFIED SOURCES & DOI IDENTIFIERS

- García, Heckman & Ronda (2023), Journal of Political Economy, Perry through late midlife (2021 NBER working paper: w29057)
- Campbell et al. (2012), Developmental Psychology, Abecedarian age-30 follow-up
- Elango, García, Heckman & Hojman (2016), NBER, early childhood education synthesis